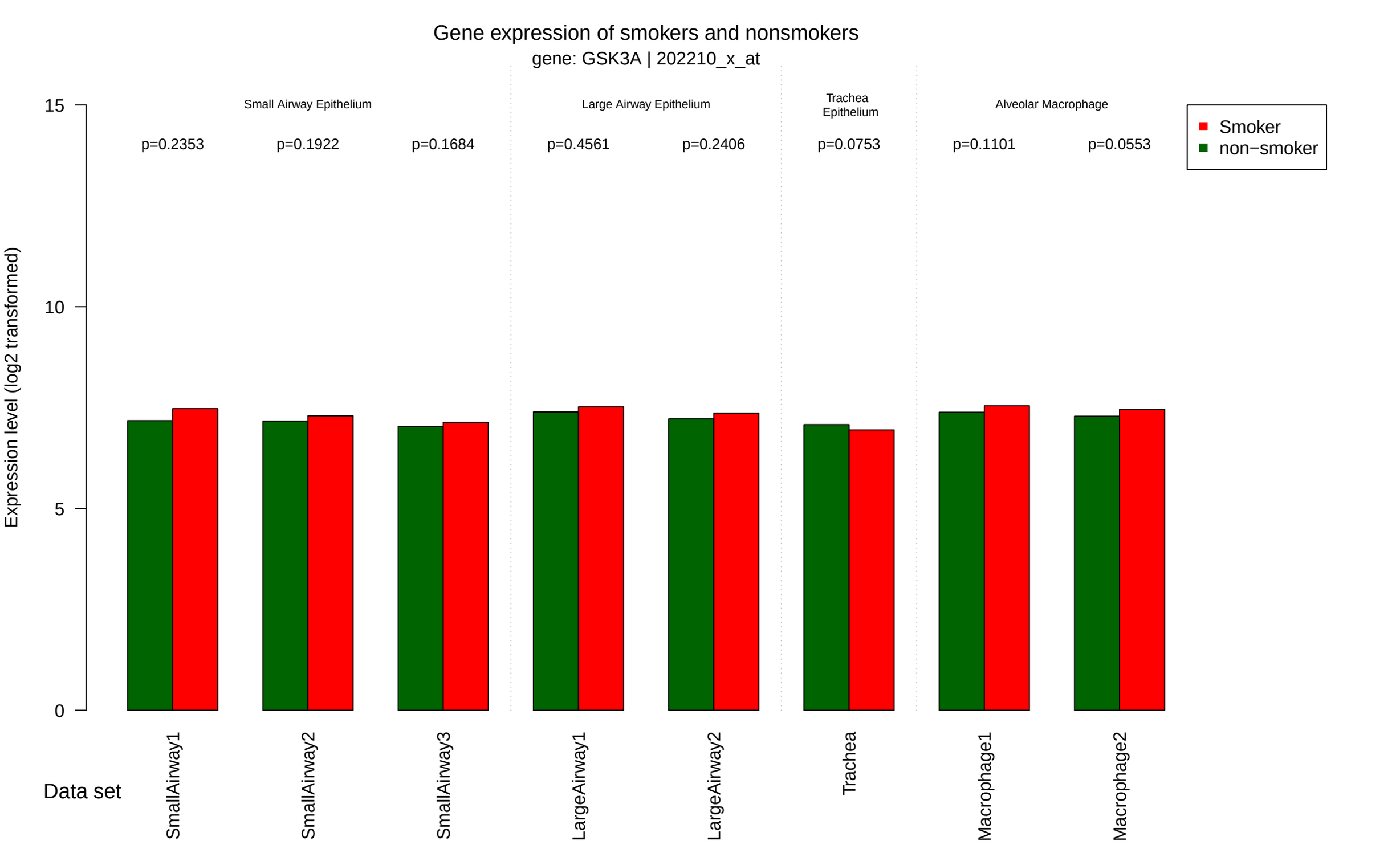
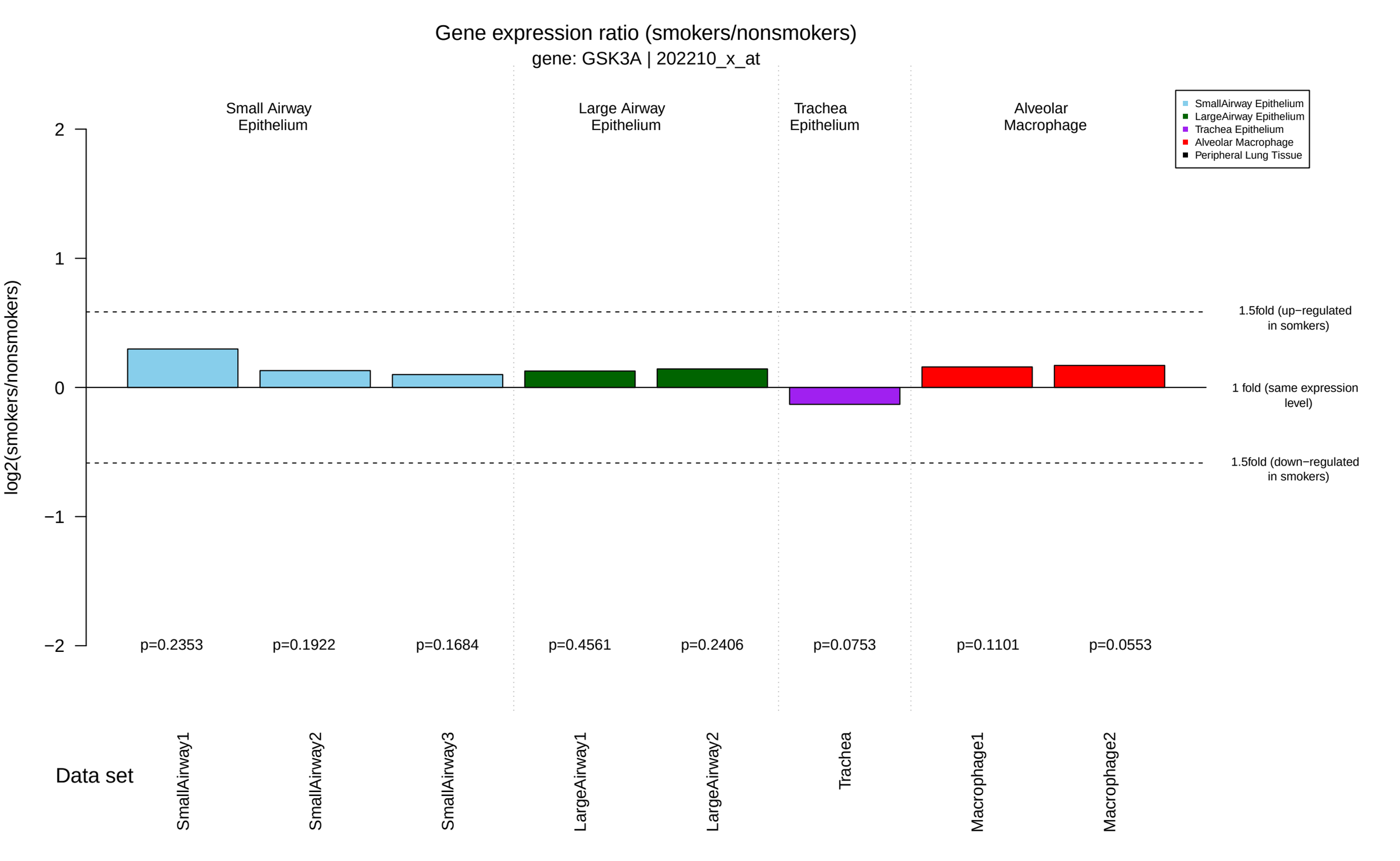
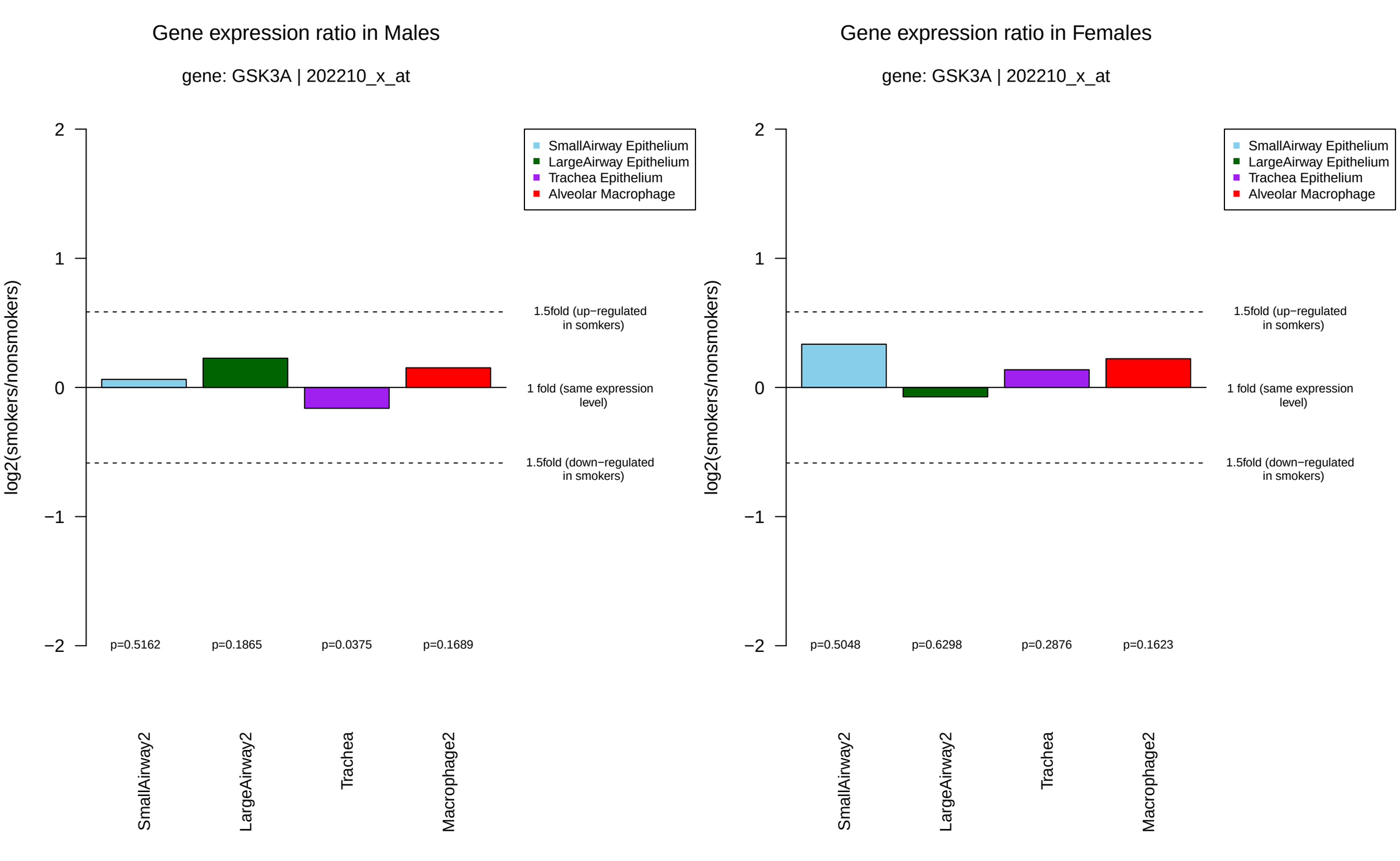
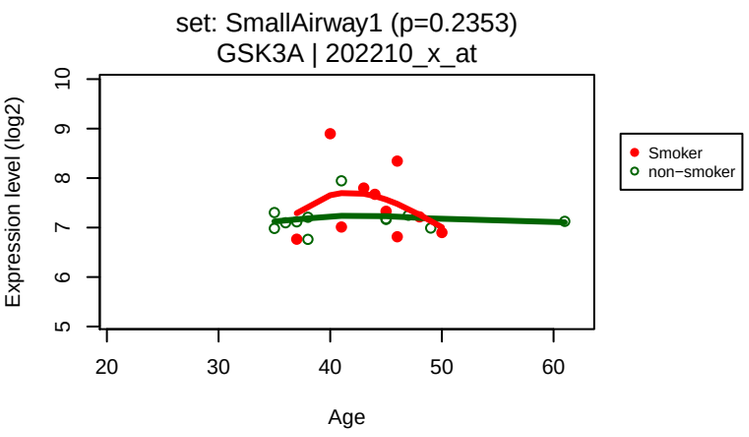
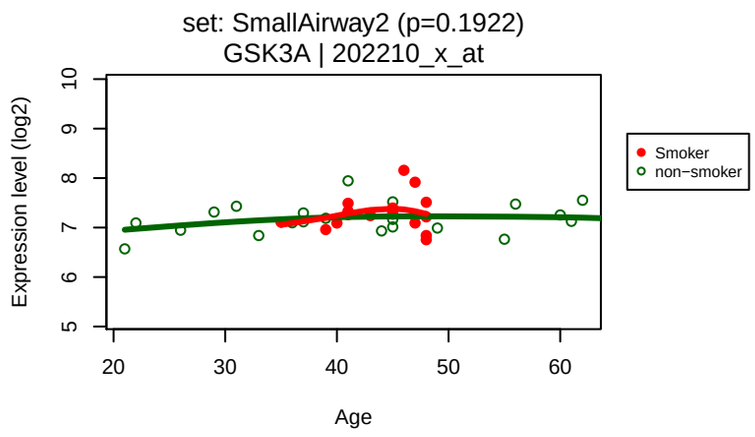
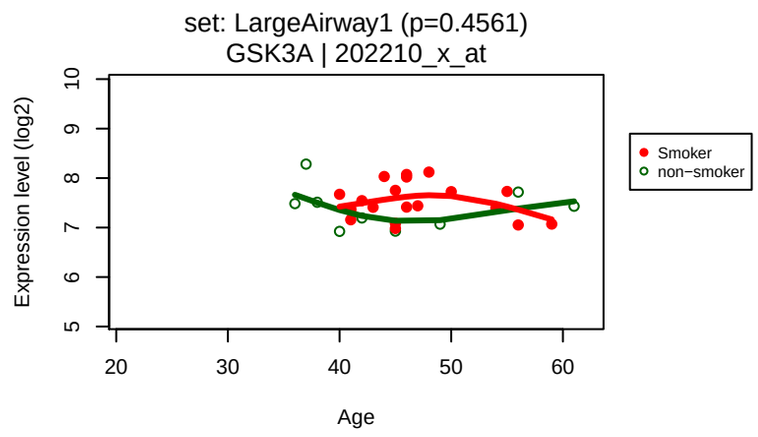

Gene expression of smokers and nonsmokers
gene: GSK3A | 202210_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2353
p=0.1922
p=0.1684
p=0.4561
p=0.2406
p=0.0753
p=0.1101
p=0.0553
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

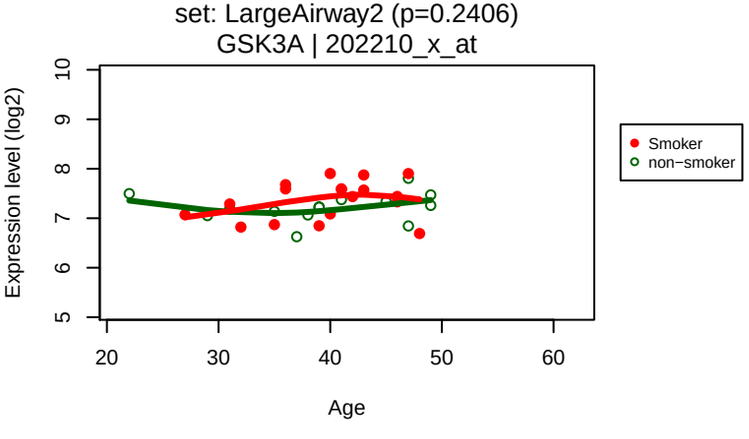

Gene expression ratio (smokers/nonsmokers)
gene: GSK3A | 202210_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2353
p=0.1922
p=0.1684
p=0.4561
p=0.2406
p=0.0753
p=0.1101
p=0.0553
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

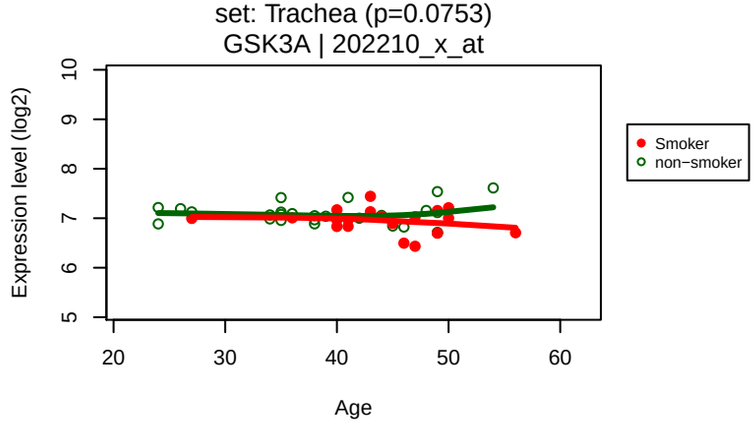

Gene expression ratio in Males
gene: GSK3A | 202210_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5162
p=0.1865
p=0.0375
p=0.1689
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSK3A | 202210_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5048
p=0.6298
p=0.2876
p=0.1623
SmallAirway2
LargeAirway2
Trachea
Macrophage2

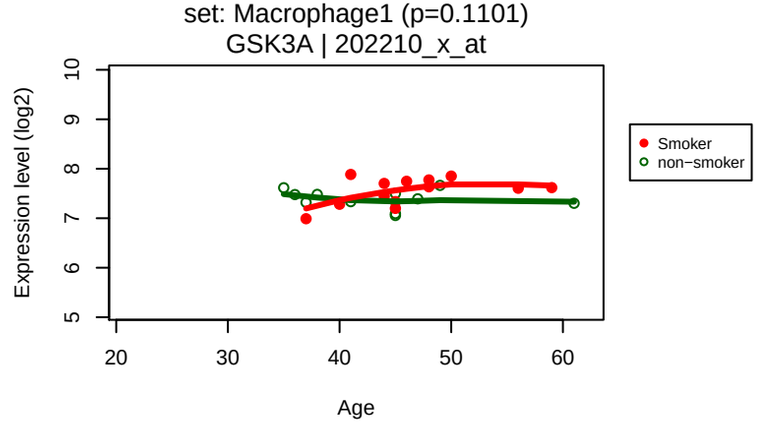

set: SmallAirway1 (p=0.2353)
GSK3A | 202210_x_at
Expression level (log2)
Age
Smoker
non−smoker

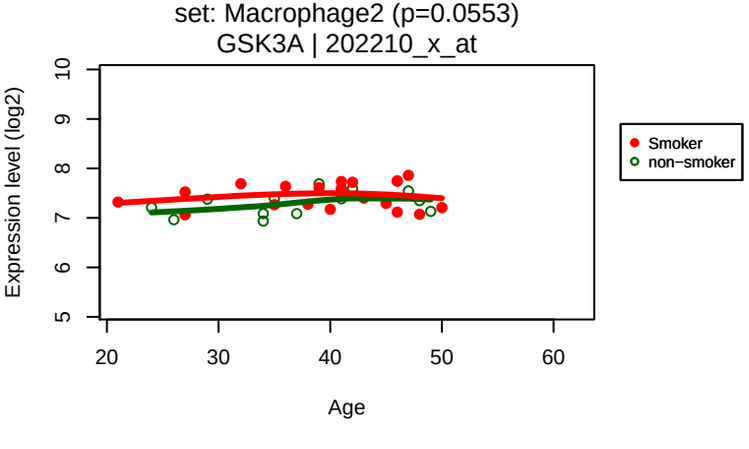

set: SmallAirway2 (p=0.1922)
GSK3A | 202210_x_at
Expression level (log2)
Age
Smoker
non−smoker

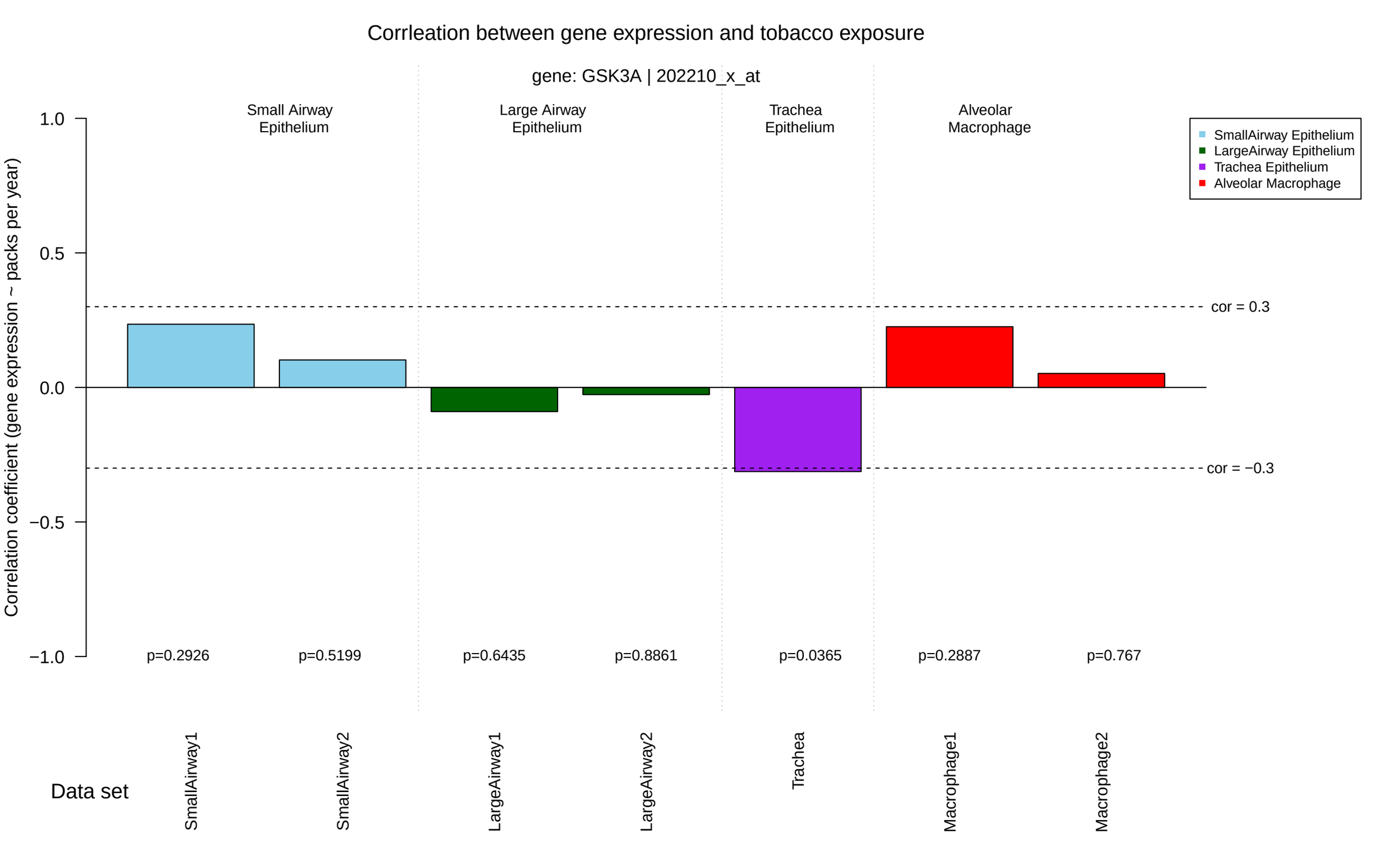

set: LargeAirway1 (p=0.4561)
GSK3A | 202210_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2406)
GSK3A | 202210_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0753)
GSK3A | 202210_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1101)
GSK3A | 202210_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0553)
GSK3A | 202210_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSK3A | 202210_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2926
p=0.5199
p=0.6435
p=0.8861
p=0.0365
p=0.2887
p=0.767
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2